AWS Interview Guide

Complete preparation guide covering compute, storage, networking, security, and real interview Q&A.

1. Compute Services

AWS compute options range from fully managed virtual machines to serverless functions and container orchestration platforms. Interviewers want to see that you can choose the right compute model for a given workload and cost profile.

EC2 (Elastic Compute Cloud)

- Resizable virtual servers billed by the second or hour.
- Pricing models: On-Demand (pay as you go), Reserved (1-3 year commitment for discount), Spot (bid on unused capacity, can be reclaimed), Savings Plans.
- Instance families are optimized for different needs: General purpose (M), Compute optimized (C), Memory optimized (R), Storage optimized (I/D), GPU (P/G).
- Auto Scaling Groups adjust instance count based on demand using launch templates.

Lambda (Serverless Compute)

- Runs code in response to events without provisioning servers; billed per invocation and execution time (ms).
- Has a maximum execution timeout (15 minutes) and memory-linked CPU allocation.
- Cold starts happen when a new execution environment must be initialized — mitigated with Provisioned Concurrency.
- Common triggers: API Gateway, S3 events, DynamoDB Streams, EventBridge, SQS.

ECS / EKS (Container Orchestration)

- ECS is AWS's native container orchestrator; EKS is managed Kubernetes.
- Fargate is the serverless launch type for both — no EC2 instances to manage.
- Task Definitions (ECS) / Pod specs (EKS) describe container images, CPU/memory, and networking.

2. Storage & Databases

Choosing the correct storage class or database engine is one of the most frequently tested AWS decisions in interviews — know the trade-offs, not just the definitions.

S3 (Simple Storage Service)

- Object storage organized into buckets; virtually unlimited scale, 99.999999999% (11 nines) durability.
- Storage classes: Standard, Intelligent-Tiering, Standard-IA, One Zone-IA, Glacier Instant Retrieval, Glacier Flexible Retrieval, Glacier Deep Archive — trade retrieval speed for cost.
- Versioning, lifecycle policies (auto-transition/expire objects), and bucket policies for access control.

EBS (Elastic Block Store)

- Persistent, network-attached block storage for a single EC2 instance (unless using Multi-Attach).
- Volume types: gp3 (general purpose SSD), io2 (high IOPS), st1/sc1 (throughput-optimized HDD for large sequential workloads).
- Snapshots are incremental backups stored in S3.

RDS & DynamoDB

- RDS: managed relational databases (MySQL, PostgreSQL, MariaDB, SQL Server, Oracle). Multi-AZ gives synchronous standby replicas for failover; Read Replicas scale reads asynchronously.
- DynamoDB: fully managed NoSQL key-value/document store with single-digit millisecond latency at any scale; supports on-demand or provisioned throughput and Global Tables for multi-region replication.

3. Networking & Security

Networking and IAM questions test whether you understand the shared responsibility model and can design a secure, isolated environment.

VPC (Virtual Private Cloud)

- A logically isolated network you define with a CIDR block, split into public and private subnets across Availability Zones.
- Internet Gateway allows public subnet resources to reach the internet; NAT Gateway lets private subnet resources initiate outbound traffic without being publicly reachable.
- Route Tables control traffic flow between subnets and gateways.

IAM (Identity and Access Management)

- Follows the principle of least privilege — grant only the permissions required.
- Users (people/apps), Groups (collections of users), Roles (temporary credentials assumable by services or federated users), and Policies (JSON documents defining permissions).
- Prefer roles over long-lived access keys wherever possible.

Security Groups vs NACLs

- Security Groups: stateful, instance-level firewall — return traffic is automatically allowed.
- NACLs (Network ACLs): stateless, subnet-level firewall — must explicitly allow both inbound and outbound rules; evaluated in rule-number order.

4. Real Interview Q&A

> **Q: What is the difference between Availability Zones and Regions?**
>
> **A:** A Region is a geographic area (e.g., us-east-1) containing multiple isolated data centers called Availability Zones (AZs). AZs within a region are connected with low-latency links but are physically separated for fault isolation. Designing across multiple AZs gives high availability; designing across multiple Regions protects against a region-wide outage and can reduce latency for global users.

> **Q: When would you choose Lambda over EC2?**
>
> **A:** Choose Lambda for short-lived, event-driven, or unpredictable workloads where you don't want to manage servers and want to pay only for execution time — e.g., processing an S3 upload or an API request. Choose EC2 when you need long-running processes, full OS control, specialized software installations, consistent high throughput, or workloads that exceed Lambda's 15-minute execution limit.

> **Q: Explain the difference between a Security Group and a NACL, and when you'd use each.**
>
> **A:** Security Groups operate at the instance/ENI level and are stateful, meaning if you allow inbound traffic, the response is automatically allowed out. NACLs operate at the subnet level, are stateless, and require explicit inbound and outbound rules. In practice, Security Groups handle most day-to-day access control; NACLs are used as a coarse, subnet-wide layer of defense, such as explicitly blocking a malicious IP range across an entire subnet.

> **Q: How does Auto Scaling work with EC2, and what triggers a scaling action?**
>
> **A:** An Auto Scaling Group maintains a desired number of EC2 instances based on a launch template and scaling policies. Scaling can be triggered by target tracking (e.g., keep average CPU at 50%), step scaling based on CloudWatch alarms, scheduled actions for predictable load patterns, or predictive scaling using machine learning on historical data.

Q: What is the difference between RDS Multi-AZ and Read Replicas?

A: Multi-AZ maintains a synchronous standby copy of the database in a different AZ purely for high availability and automatic failover — it is not used for read traffic. Read Replicas are asynchronous copies (in the same or different region) that can serve read-only queries to scale read-heavy workloads, and can also be promoted to a standalone database if needed.

Q: How would you design a highly available, scalable web application on AWS?

A: Place an Application Load Balancer in front of an Auto Scaling Group of EC2 instances (or Fargate tasks) spread across at least two AZs. Use RDS Multi-AZ (or DynamoDB) for the database layer, S3 with CloudFront for static assets, and Route 53 for DNS with health checks. Store secrets in Secrets Manager, and use CloudWatch for monitoring and alarms that drive the scaling policies.

Q: What are the S3 storage classes and how do you decide which to use?

A: S3 Standard suits frequently accessed data; Standard-IA and One Zone-IA suit infrequently accessed but rapidly retrievable data; Intelligent-Tiering automatically moves objects between tiers based on access patterns; Glacier Instant/Flexible Retrieval and Glacier Deep Archive suit long-term archival where retrieval can take minutes to hours. The decision balances access frequency, retrieval latency requirements, and cost.

Q: What is the difference between an IAM Role and an IAM User?

A: An IAM User represents a permanent identity (person or application) with long-term credentials. An IAM Role has no long-term credentials; it is assumed temporarily by a trusted entity (an EC2 instance, a Lambda function, another AWS account, or a federated identity) and issues short-lived credentials via STS, which is more secure than embedding access keys.

Q: What happens during a Lambda cold start and how can you reduce its impact?

A: A cold start occurs when Lambda has to provision a new execution environment — downloading the code package, initializing the runtime, and running any code outside the handler — before it can process the first invocation. It adds latency, especially for languages with heavier runtime initialization (like Java) or larger deployment packages. Mitigations include Provisioned Concurrency (pre-warmed environments), reducing package size, minimizing dependencies loaded at init time, and choosing lighter runtimes for latency-sensitive paths.

Q: What is the shared responsibility model in AWS?

A: AWS is responsible for the security 'of' the cloud — physical infrastructure, hardware, networking, and the virtualization layer. The customer is responsible for security 'in' the cloud — data encryption, IAM configuration, OS patching (for EC2), network access control, and application-level security. The exact split shifts depending on the service (e.g., Lambda pushes more responsibility to AWS than EC2 does).

Q: How does a NAT Gateway differ from an Internet Gateway?

A: An Internet Gateway allows resources in a public subnet to send and receive traffic directly to and from the internet. A NAT Gateway, placed in a public subnet, allows resources in a private subnet to initiate outbound connections to the internet (e.g., for software updates) while preventing unsolicited inbound connections from reaching them.

Pro Tips for the Interview

- Be ready to draw or describe a simple VPC diagram (public/private subnets, IGW, NAT, route tables) — this is one of the most common whiteboard questions.
- Know the cost trade-offs, not just the feature list — interviewers often probe 'why not just use X instead' to test judgment.
- For any service, be ready to state its high-availability story (Multi-AZ, cross-region replication, etc.).
- Practice explaining IAM least-privilege with a concrete example — e.g., a Lambda function that only needs read access to one S3 bucket.